

DSC 335, Handout 09: Posterior Approximation (*Chapter 6-7*)

LEARNING OBJECTIVES.

- *Justify the analytical approach to finding the posterior in conjugate families using simulations.*
- *Conduct grid approximation and Markov Chain Monte Carlo (MCMC) simulations to approximate posterior distributions.*

As we have seen, there are a few specific scenarios where we are able to mathematically specify the posterior distribution. But when dealing with more complicated models, this will no longer be the case. Instead, we must learn to approximate the posterior distributions.

Grid Approximation

Markov Chain Monte Carlo (MCMC)

RStan Commands

In the defining the model step:

- **data:**
- **parameters:**
- **model:**

In the simulation step, in the `stan` function:

- **model code:**
- **data:**
- **chains:**
- **iter:**
- **seed:**

Markov Chain Diagnostics

Motivating questions: Markov chains mimic a random sample that eventually converges to the posterior approximation. But how do we know if the chain has converged? And how do we know if the approximation is reasonable?

Trace Plots

Comparing Parallel Chains

Autocorrelation Plots

Effective sample size ratio

R-hat

Metropolis-Hastings algorithm

A Metropolis-Hastings Markov chain for the posterior $f(\mu|y)$, $\{\mu^{(1)}, \dots, \mu^{(n)}\}$ moves from $\mu^{(i)}$ to $\mu^{(i+1)}$ through the following process